Data Collection and Preprocessing Phase

Date	15 August 2026
Project Title	AI Career Pulse: Decoding the Job Market

Data Quality Report

Ahead of building the Tableau dashboard, the AI job-market dataset was put through a structured quality review during the preliminary phase. The assessment below covers all 1,500 records in the dataset.

Data Source	Data Quality Issue	Severity	Resolution Plan
AI Job-Market Dataset	Completeness check: across all 25 fields in the AI job-market file, not a single cell was found to be empty.	Low	Since every field is fully populated, no imputation or missing-value treatment is needed for this dataset.
AI Job-Market Dataset	Duplicate check: scanning the full record set turned up zero duplicate rows and zero repeated job_id values.	Low	As no duplicate entries exist, there is nothing to remove from this dataset on that front.
AI Job-Market Dataset	Experience-field mismatch: 1,099 of the 1,500 records show experience_level tags that don't line up with the years_of_experience value. AIJOB0002, for instance, is marked 'Senior (6-9 yrs)' despite showing only 2 years of experience.	High	Derive experience_level directly from years_of_experience going forward, using a standard banding — 0-2 Entry, 3-5 Mid, 6-9 Senior, 10+ Lead — and re-tag every record against that rule. The same years-of-experience figure should also be used to check and correct any other experience-related flags.
AI Job-Market Dataset	Salary-figure mismatch: in 288 records (19.2% of the dataset), annual_salary_usd comes in above salary_max_usd. AIJOB0003 is one example, listing an annual_salary_usd of $360,000 against a salary_max_usd of $300,000.	Moderate	Confirm with the data source whether annual_salary_usd and salary_min_usd/salary_max_usd are meant to represent the same figure, then apply a single, documented rule to reconcile the two before these fields are used in Tableau.
AI Job-Market Dataset	Salary-tier mismatch: 36 records carry an annual_salary_usd that falls outside the bracket implied by their assigned salary_tier — for example, several $100,000	Moderate	Fix a single set of salary-tier boundaries and re-derive salary_tier for every record from annual_salary_usd using that rule, before this field feeds into the

	postings are still tagged 'Entry (<$100k)'.		Tableau dashboard.
AI Job-Market Dataset	Range check: demand_score falls between 68 and 98, benefits_score_10 between 6.0 and 9.8, posting_month between 1 and 12, and posting_year between 2025 and 2026 — all values sit inside these expected bounds.	Low	These fields all fall within their expected bounds, so no corrective action is needed here.

Data Cleaning Process

Cleaning Area	Cleaning Process
Missing values	All 25 fields were checked for empty cells that might need treatment or imputation; none were found at this stage.
Duplicates	Every field was scanned for duplicated entries; no duplicated records turned up at this stage.
Experience fields	Cross-check experience_level against years_of_experience and re-standardize experience_level wherever the two disagree.
Salary fields	Confirm that annual_salary_usd lines up with salary_min_usd/salary_max_usd, and clarify the source-side rule wherever it does not.
Salary tier	Recalculate and standardize salary_tier from annual_salary_usd using a single documented rule, as described in the field definitions.
Final validation	Re-check all fields once corrections are applied and confirm no outstanding data-quality issues remain before the dataset moves into Tableau.